



Airbnb Pricing Prediction in NYC

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Objective & Value Proposition

Objective:

- ❑ Investigate what affects Airbnb listing price in NYC.
- ❑ Build predictive models that use listing features to estimate price

Value:

Understanding the dynamics could offer value to a variety of stakeholders.

- ❑ *Hosts*: price their properties better and optimize their offerings to increase visibility and bookings.
- ❑ *Tourists*: learn which feature influences rental pricing.
- ❑ *Airbnb*: understand how users interact with listings based on property features and listing price.



Dataset Used

- ❑ A publicly available New York City Airbnb listing dataset:
<https://www.kaggle.com/datasets/arianazmoudeh/airbnbopendata/data>
- ❑ **Size:** 100,000+ rows, 26 features.
- ❑ **Context:** New York City Airbnb listings – rich data for pricing, tourism, and economic behavior analysis.
- ❑ **Key Features:** Neighbourhood, Instant Bookable, Cancellation Policy, Room Type, Construction Year, Price, Service Fee, Minimum Nights, Number of Reviews, etc.

#	Column	Non-Null	Count	Dtype
0	id	102058	non-null	int64
1	name	101808	non-null	object
2	host_id	102058	non-null	int64
3	host_identity_verified	101769	non-null	object
4	host_name	101654	non-null	object
5	neighbourhood_group	102029	non-null	object
6	neighbourhood	102042	non-null	object
7	lat	102050	non-null	float64
8	long	102050	non-null	float64
9	country	101526	non-null	object
10	country_code	101927	non-null	object
11	instant_bookable	101953	non-null	object
12	cancellation_policy	101982	non-null	object
13	room_type	102058	non-null	object
14	construction_year	101844	non-null	float64
15	price	101811	non-null	object
16	service_fee	101785	non-null	object
17	minimum_nights	101658	non-null	float64
18	number_of_reviews	101875	non-null	float64
19	last_review	86226	non-null	object
20	reviews_per_month	86240	non-null	float64
21	review_rate_number	101739	non-null	float64
22	calculated_host_listings_count	101739	non-null	float64
23	availability_365	101610	non-null	float64
24	house_rules	50216	non-null	object
25	license	2	non-null	object

Dataset Info

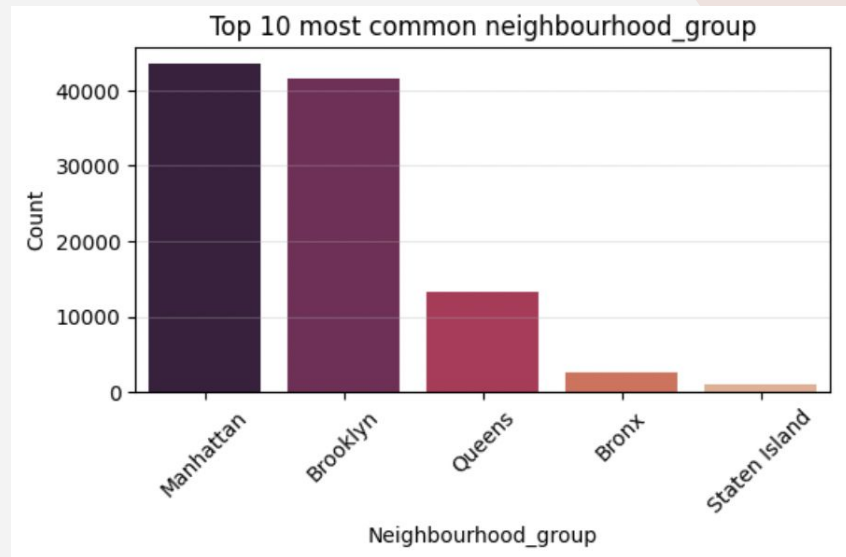
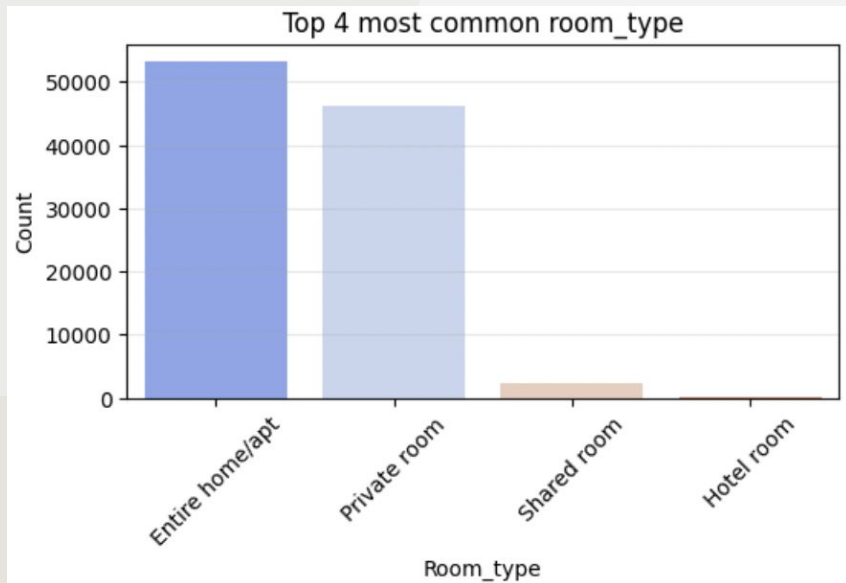


Dataset Used (Cont'd)

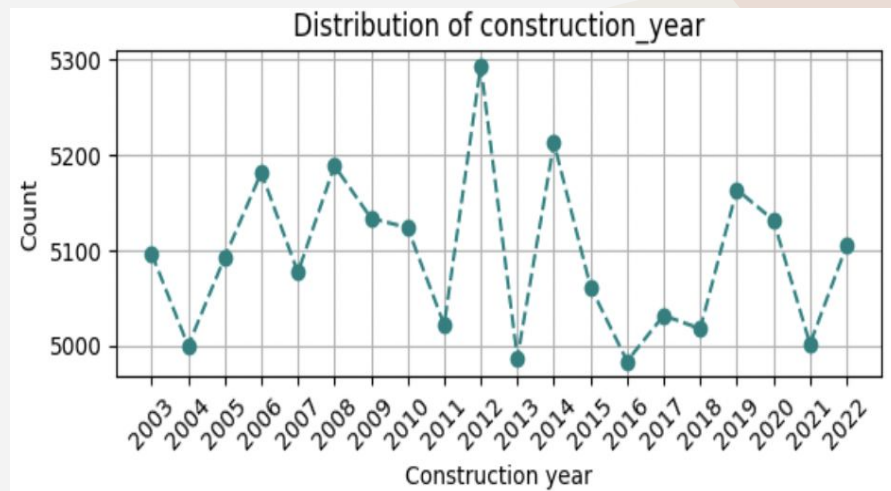
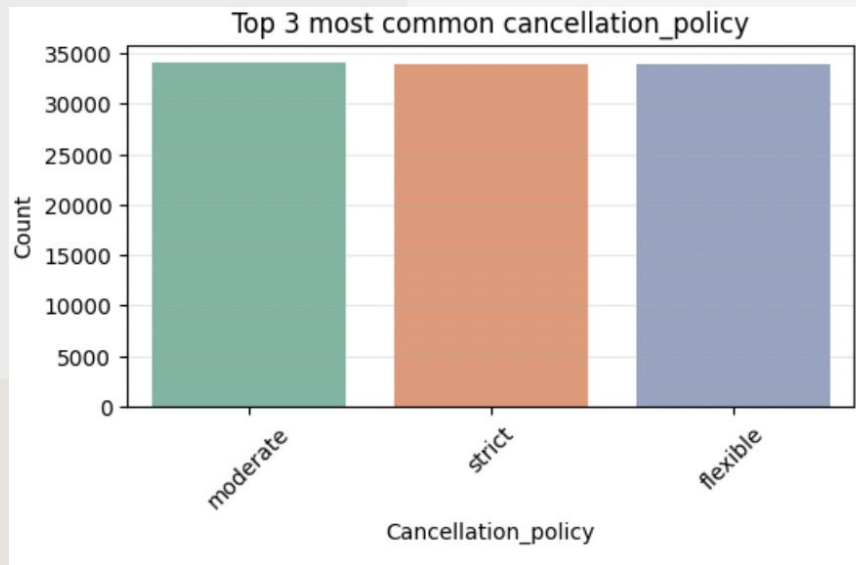
Data Cleaning:

- ❑ Drop duplicate rows
- ❑ Deal with NaN values in each columns
 - ❑ Drop columns that are not useful for our analysis
 - ❑ Drop columns with too many NaNs
 - ❑ Fill NaN values with median or mean, depending on the scenarios
- ❑ Convert data into more standard forms
 - ❑ Remove dollar sign
 - ❑ Fix typo

Major Learnings from EDA



Major Learnings from EDA (Cont'd)



Major Learnings from EDA (Cont'd)

Distribution of Airbnb Listings and Major Tourist Attractions in NYC
Hover on the map: red dots show housing prices, blue dots show tourist attractions.
Use the plus (+) and minus (-) buttons on the upper right to zoom in and out of the map.

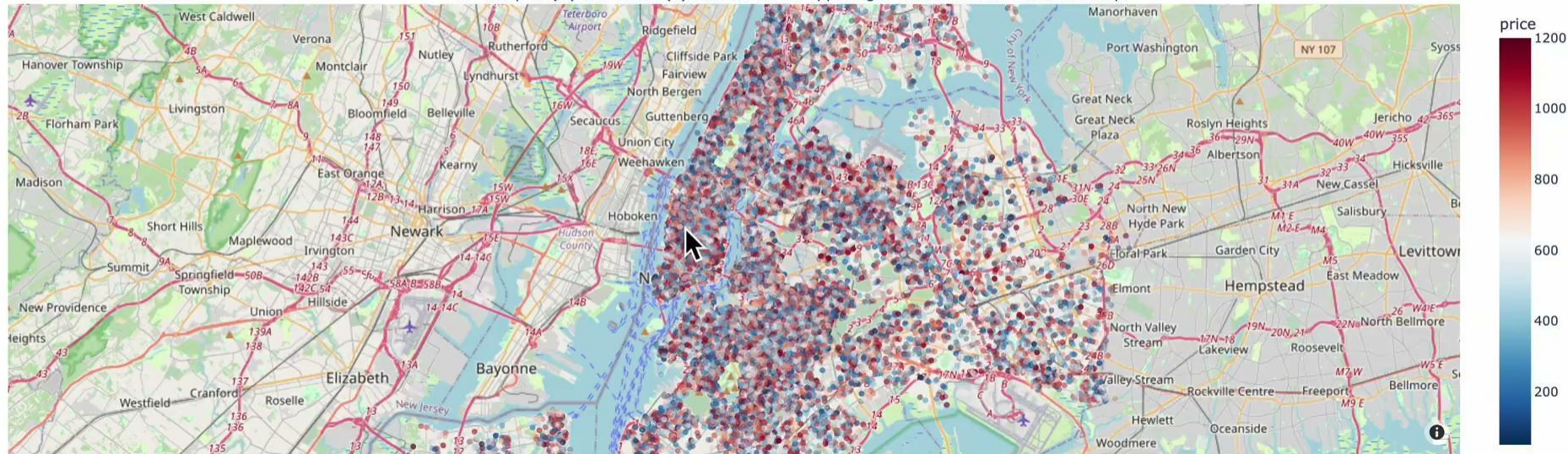


Major Learnings from EDA (Cont'd)

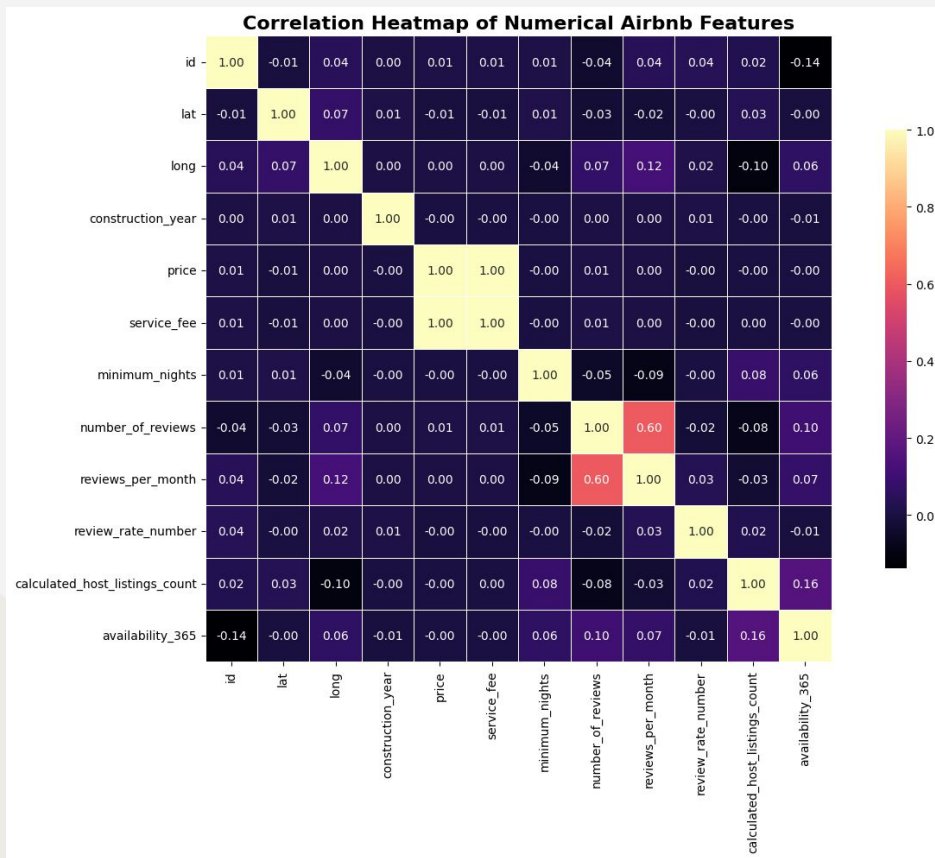
Airbnb Price Distribution Across NYC

Hover on the map: red dots show higher housing prices, blue dots show lower housing prices.

Use the plus (+) and minus (-) buttons on the upper right to zoom in and out of the map.



Major Learnings from EDA (Cont'd)



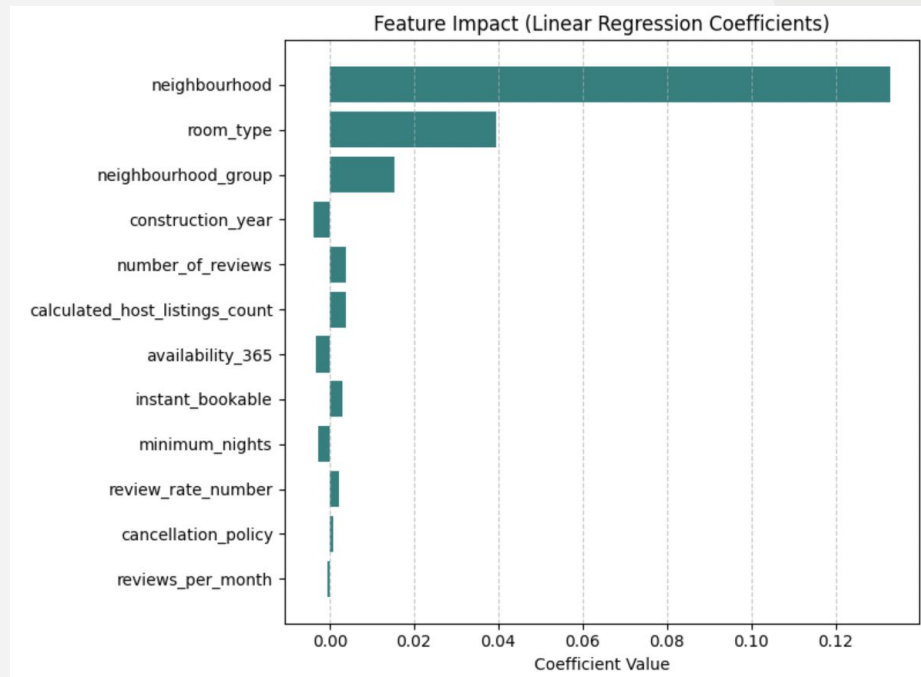
Modeling Results

Baseline Model: Linear Regression

- ❑ Log_Price vs. other 12 features

Model Assessment:

- ❑ Root Mean Squared Error: 0.7293682
 - ❑ The model's predictions are off by about 0.73 on average in the log-transformed scale of the price.
- ❑ R-squared: -0.0013013
 - ❑ Model performs worse than just predicting the mean price for all listings.



Modeling Results (Cont'd)

Tree-based Model 1: Gradient Boosted Trees

Baseline Model Assessment:

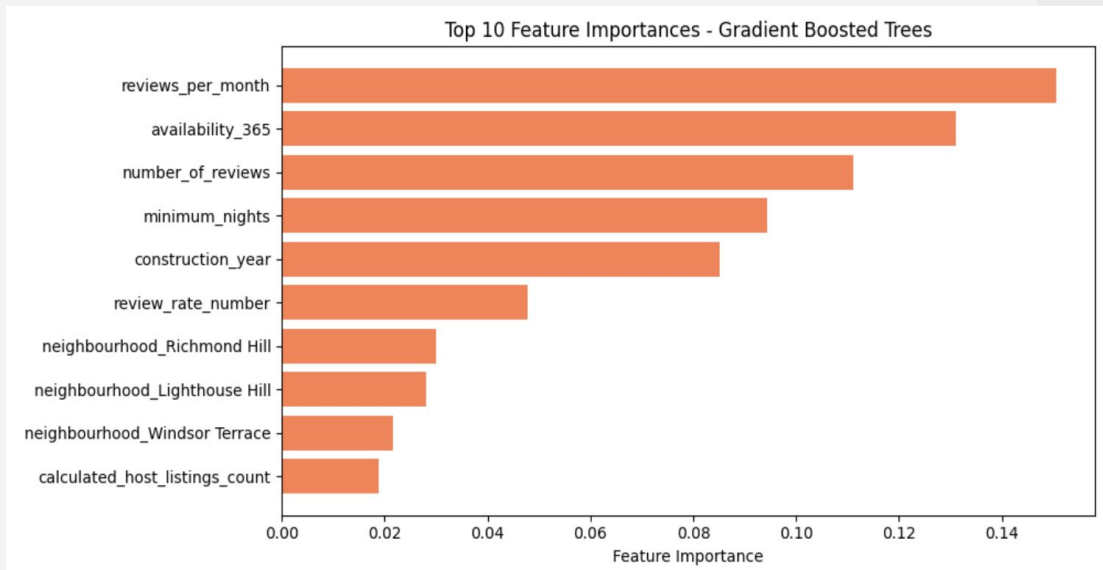
- ❑ RMSE: 0.7278230
- ❑ R^2 : 0.0029368

With Model Tuning:

N_estimators: 100 → 500

Learning_rate: 0.1 → 0.01

- ❑ RMSE: 0.7283721
- ❑ R^2 : 0.0014319



Modeling Results (Cont'd)

Tree-based Model 2: Random Forest Regression

Baseline Model Assessment:

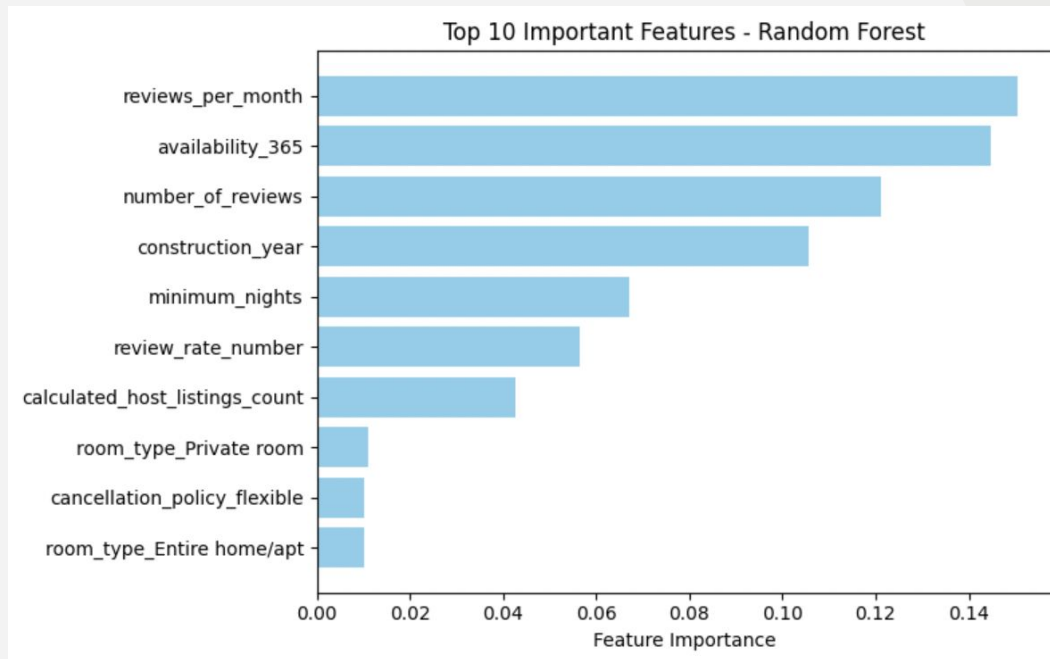
- ❑ RMSE: 0.7268015
- ❑ R^2 : 0.0057336

With Model Tuning:

N_estimators: 20 → 100

Max_depth: 10 → 15

- ❑ RMSE: 0.7229155
- ❑ R^2 : 0.0163373

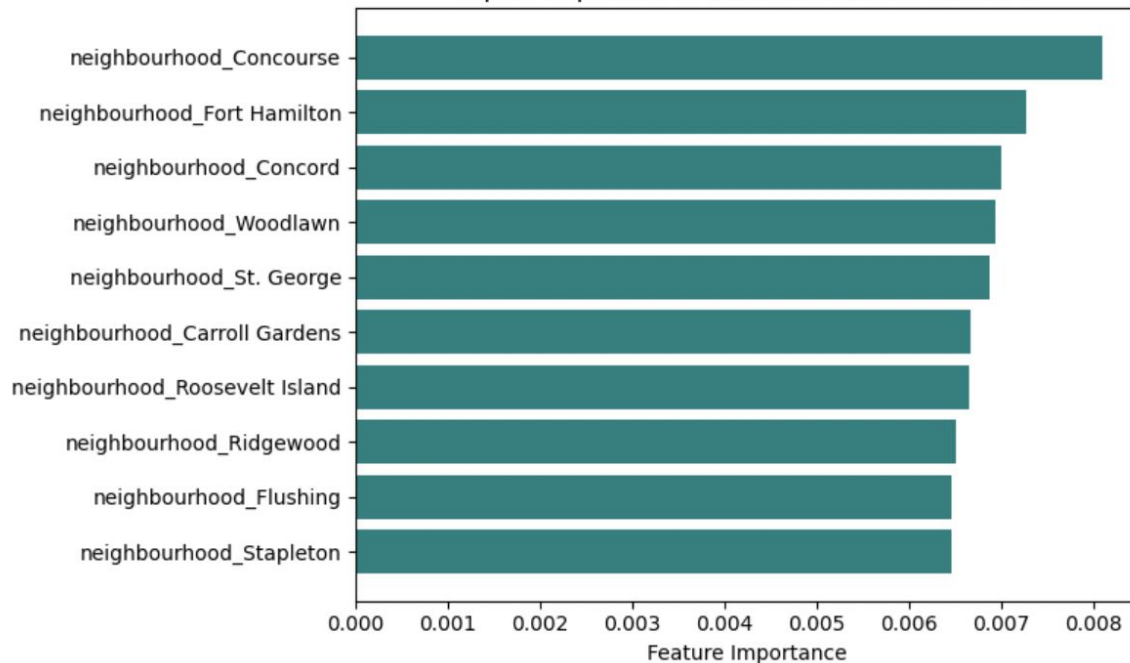


Modeling Results (Cont'd)

Tree-based
regressor
regressor
regressor
regressor
regressor

RMSE
R²: 0

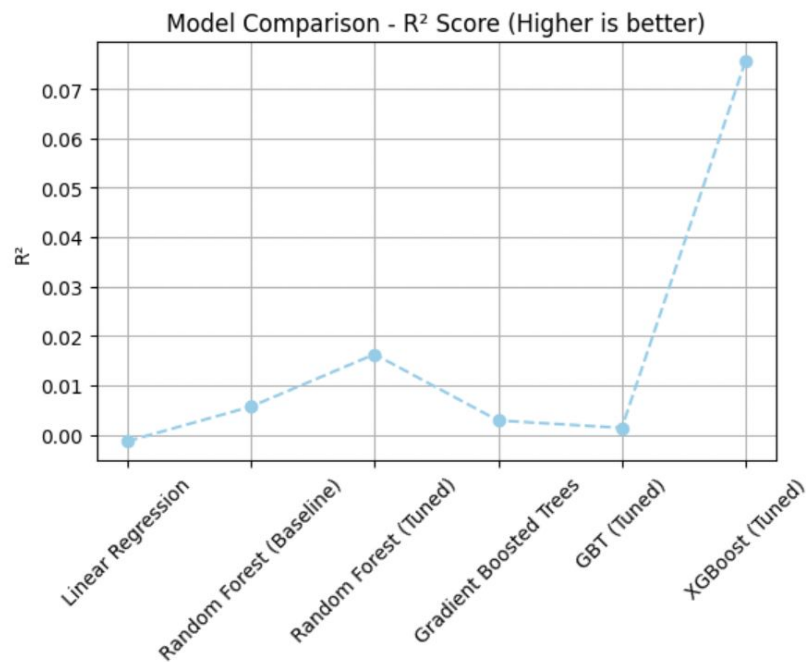
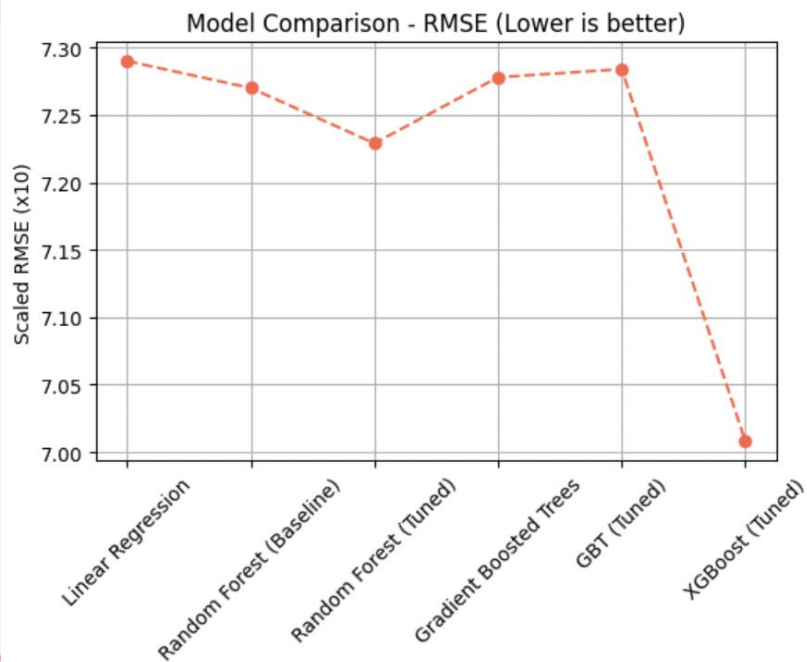
Top 10 Important Features from XGBoost Model



Result

Modeling Results (Cont'd)

Model Results Comparison



Modeling Results (Cont'd)

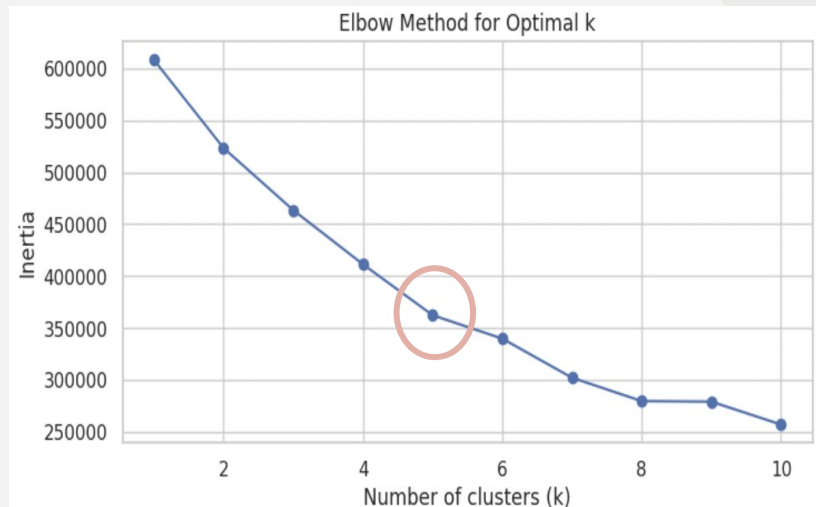
Unsupervised learning: K-Means clustering

❑ Goal:

To uncover hidden patterns in Airbnb listings by grouping them based on pricing, reviews, room type, and other features.

❑ Choosing the Number of Clusters:

- ❑ We used the Elbow Method to choose the number of clusters
- ❑ Settled on 5 clusters, balancing model complexity and interpretability.



Modeling Results (Cont'd)

Unsupervised learning: K-Means clustering

Cluster	Possible Type	Key Traits
0	Balanced Popular Listings	Short stays, high reviews, decent availability, competitive price
1	High Availability, Low Demand	Open year-round, long stays, fewer bookings
2	Outlier / Legacy Listings	Impractical min. nights, low engagement, high price
3	Budget Listings	Lowest prices, moderate reviews, limited availability
4	Seasonal Premiums	High review quality, high price, very limited availability

Hypothesis Test Results

Hypothesis Test 1:

- Do instant bookable listings tend to have higher prices than non-instant bookable ones?
 - **Null Hypothesis (H_0)**:
There is **no difference** in mean price between instant bookable and non-instant bookable listings.
 - **Alternative Hypothesis (H_1)**:
Instant bookable listings have higher mean prices.

Hypothesis Test Results (Cont'd)

Methodology:

- ❑ Use the **difference in mean log-transformed price** between instant bookable and non-instant bookable listings.
- ❑ Apply **bootstrapping** (resampling 10,000 times) to generate a sampling distribution under the null hypothesis.

Results:

- ❑ Observed difference in means: 0.0009
- ❑ P-value: 0.4178

Interpretation:

- ❑ The p-value is much greater than 0.05 → we **fail to reject** the null hypothesis.

Hypothesis Test Results (Cont'd)

Hypothesis Test 2:

- Do listings with more reviews have higher prices than listings that have fewer reviews?
- **Null Hypothesis (H_0):**
There is **no difference** in price between listings with many reviews and few reviews.
- **Alternative Hypothesis (H_1):**
Listings with more reviews tend to have higher prices.

Hypothesis Test Results (Cont'd)

Methodology:

- ❑ Split listings into two groups: **above-median** and **below-median** number of reviews.
- ❑ Compare the **mean log_price** between the two groups.
- ❑ Use **bootstrapping** to estimate the sampling distribution and compute a p-value.

Results:

- ❑ Observed difference in log_price: 0.0078
- ❑ P-value: 0.0437

Interpretation:

- ❑ Reject the null hypothesis at the 5% significance level.

Insights & Implications

- ❑ **Identified 5 distinct listing personas**, offering a blueprint for Airbnb to better segment its host base or tailor onboarding strategies.
- ❑ **Listings with more reviews tend to have higher prices**, confirming the value of social proof in user behavior.
- ❑ **Hosts**: optimizing listing price by understanding that neighborhood and room type carry the most weight in pricing
- ❑ **Travelers**: identify affordable and high-density areas and to choose listings that balance price, availability, and review history.
- ❑ **Airbnb**: recommend pricing strategies or marketing tactics to hosts, particularly for underperforming segments like high-availability but low-review listings.

Limitations & Future Work

- ❑ **Low Predictive Power:** Even after hyperparameter tuning, our best model result still have R^2 under 0.08.
- ❑ **Data limitations:** Important variables such as listing photos and decoration aesthetics are not captured.
- ❑ **Use NLP and Computer Vision Techniques**
- ❑ **Enrich location data** such as distance to public transit, local amenities, or tourist attractions, to better contextualize price differences.
- ❑ **Try causal methods** and move beyond prediction to understand cause-effect relationships

THANKS!

Have questions?

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